

Redes de Computadoras – Parcial 2

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A: $23 + 1 = 24$

B: $26 + 1 = 27$

Red base asignada (Ejercicio 1): 10.24.27.0 / 24

Red base asignada (Ejercicio 2): 192.24.27.0 / 24

Ejercicio 1

Configuración de interfaces y tablas de rutas en routers:

The screenshot shows the GNS3 interface with a terminal window for router R1. The terminal displays the following configuration commands and output:

```
changed state to down
R1(config)#ip route 10.24.27.112 255.255.255.248 10.24.27.122
R1(config)#ip route 10.24.27.124 255.255.255.252 10.24.27.122
R1(config)#end
R1#
Building configuration...
*Mar 1 00:06:55.315: %SYS-5-CONFIG_I: Configured from console by console[OK]
R1#
*Mar 1 00:08:33.247: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0,
changed state to up
R1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 10.24.27.1      YES manual up          up
FastEthernet0/1 10.24.27.65     YES manual up          up
FastEthernet1/0 10.24.27.97     YES manual up          up
Serial2/0       10.24.27.121    YES manual up          up
Serial2/1       unassigned      YES unset  administratively down down
Serial2/2       unassigned      YES unset  administratively down down
Serial2/3       unassigned      YES unset  administratively down down
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

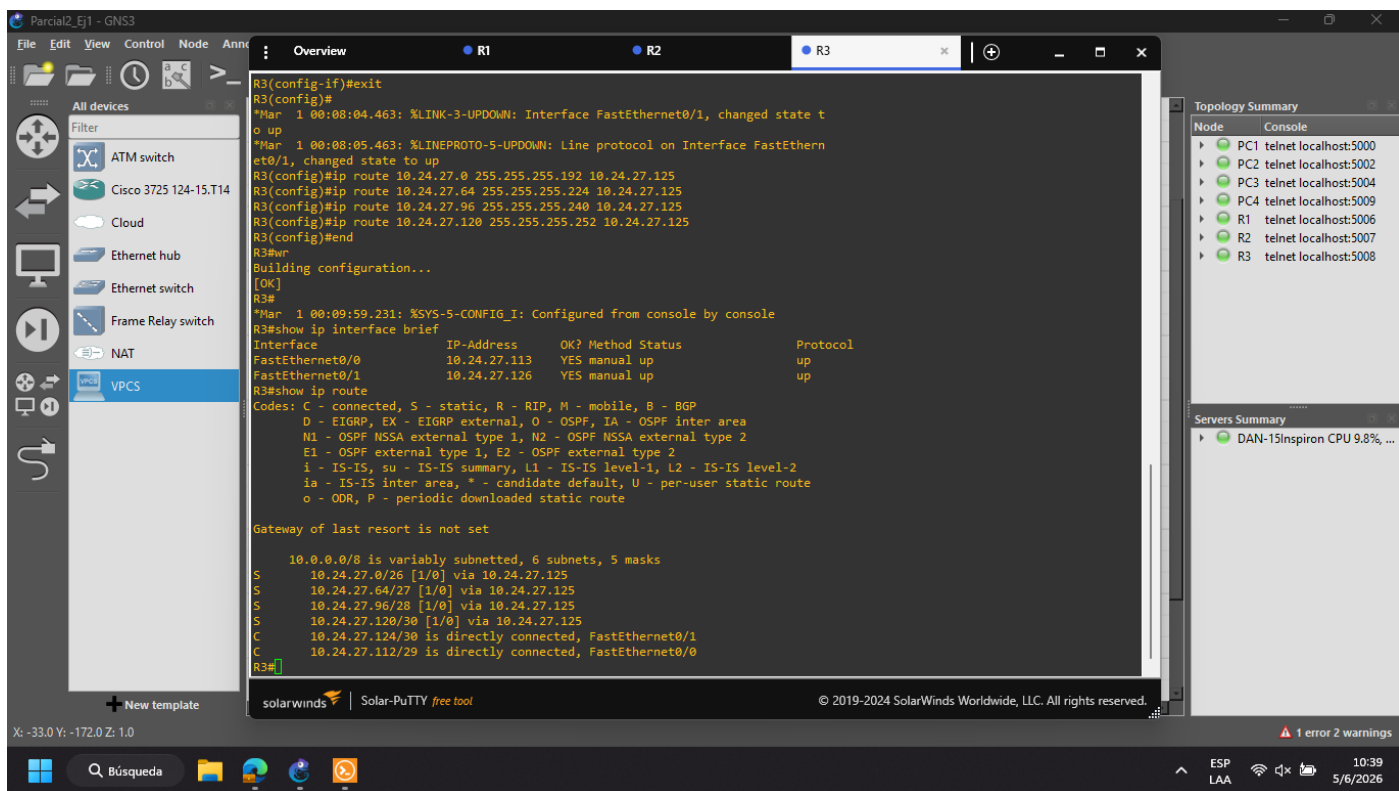
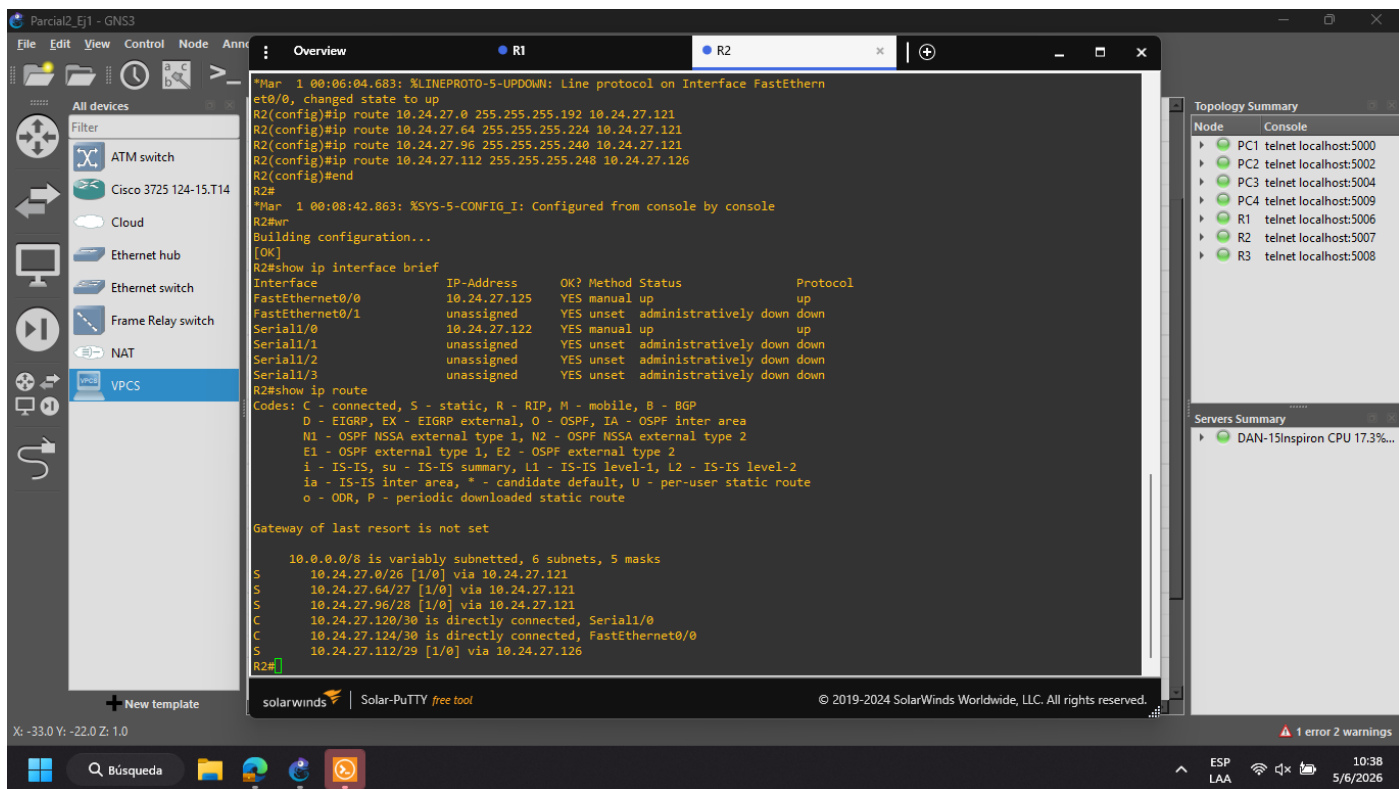
Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
C    10.24.27.0/26 is directly connected, FastEthernet0/0
C    10.24.27.64/27 is directly connected, FastEthernet0/1
C    10.24.27.96/28 is directly connected, FastEthernet1/0
C    10.24.27.120/30 is directly connected, Serial2/0
S    10.24.27.124/30 [1/0] via 10.24.27.122
S    10.24.27.112/29 [1/0] via 10.24.27.122
R1#
```

The right sidebar shows the Topology Summary with the following nodes:

Node	Console
PC1	telnet localhost:5000
PC2	telnet localhost:5002
PC3	telnet localhost:5004
PC4	telnet localhost:5009
R1	telnet localhost:5006
R2	telnet localhost:5007
R3	telnet localhost:5008

The Servers Summary shows: DAN-15Inspiron CPU 20.3%...



Configuración de IPs y pruebas de ping/trace entre PCs:

Parcial2_Ej1 - GNS3

File Edit View Control Node Annotations

Overview PC1

PC1> show ip

```
NAME       : PC1[1]
IP/MASK    : 10.24.27.2/26
GATEWAY    : 10.24.27.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 10032
RHOST:PORT : 127.0.0.1:10033
MTU        : 1500
```

PC1> ping 10.24.27.66

```
10.24.27.66 icmp_seq=1 timeout
84 bytes from 10.24.27.66 icmp_seq=2 ttl=63 time=31.461 ms
84 bytes from 10.24.27.66 icmp_seq=3 ttl=63 time=30.887 ms
```

PC1> ping 10.24.27.98

```
10.24.27.98 icmp_seq=1 timeout
84 bytes from 10.24.27.98 icmp_seq=2 ttl=63 time=30.809 ms
84 bytes from 10.24.27.98 icmp_seq=3 ttl=63 time=30.843 ms
```

PC1> ping 10.24.27.114

```
10.24.27.114 icmp_seq=1 timeout
10.24.27.114 icmp_seq=2 timeout
84 bytes from 10.24.27.114 icmp_seq=3 ttl=61 time=90.541 ms
84 bytes from 10.24.27.114 icmp_seq=4 ttl=61 time=91.234 ms
```

PC1>

Topology Summary

Node	Console
PC1	telnet localhost:5000
PC2	telnet localhost:5002
PC3	telnet localhost:5004
PC4	telnet localhost:5009
R1	telnet localhost:5006
R2	telnet localhost:5007
R3	telnet localhost:5008

Servers Summary

DAN-15Inspiron CPU 19.1%...

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X: 0.0 Y: 0.0 Z: 0.5

1 error 2 warnings

10:41 5/6/2026

Parcial2_Ej1 - GNS3

File Edit View Control Node Annotations

Overview PC1

MAC : 00:50:79:66:68:00
LPORT : 10032
RHOST:PORT : 127.0.0.1:10033
MTU : 1500

PC1> ping 10.24.27.66

```
10.24.27.66 icmp_seq=1 timeout
84 bytes from 10.24.27.66 icmp_seq=2 ttl=63 time=31.461 ms
84 bytes from 10.24.27.66 icmp_seq=3 ttl=63 time=30.887 ms
```

PC1> ping 10.24.27.98

```
10.24.27.98 icmp_seq=1 timeout
84 bytes from 10.24.27.98 icmp_seq=2 ttl=63 time=30.809 ms
84 bytes from 10.24.27.98 icmp_seq=3 ttl=63 time=30.843 ms
```

PC1> ping 10.24.27.114

```
10.24.27.114 icmp_seq=1 timeout
10.24.27.114 icmp_seq=2 timeout
84 bytes from 10.24.27.114 icmp_seq=3 ttl=61 time=90.541 ms
84 bytes from 10.24.27.114 icmp_seq=4 ttl=61 time=91.234 ms
```

PC1> trace 10.24.27.66

```
trace to 10.24.27.66, 8 hops max, press Ctrl+C to stop
 1 10.24.27.1 15.912 ms 15.762 ms 16.239 ms
 2 *10.24.27.66 30.626 ms (ICMP type:3, code:3, Destination port unreachable)
```

PC1> trace 10.24.27.98

```
trace to 10.24.27.98, 8 hops max, press Ctrl+C to stop
 1 10.24.27.1 15.385 ms 15.173 ms 15.468 ms
 2 *10.24.27.98 30.253 ms (ICMP type:3, code:3, Destination port unreachable)
```

PC1> trace 10.24.27.114

```
trace to 10.24.27.114, 8 hops max, press Ctrl+C to stop
 1 10.24.27.1 15.456 ms 15.737 ms 15.657 ms
 2 10.24.27.122 45.772 ms 46.586 ms 44.544 ms
 3 10.24.27.126 75.696 ms 74.444 ms 76.172 ms
 4 *10.24.27.114 91.205 ms (ICMP type:3, code:3, Destination port unreachable)
```

PC1>

Topology Summary

Node	Console
PC1	telnet localhost:5000
PC2	telnet localhost:5002
PC3	telnet localhost:5004
PC4	telnet localhost:5009
R1	telnet localhost:5006
R2	telnet localhost:5007
R3	telnet localhost:5008

Servers Summary

DAN-15Inspiron CPU 9.8%...

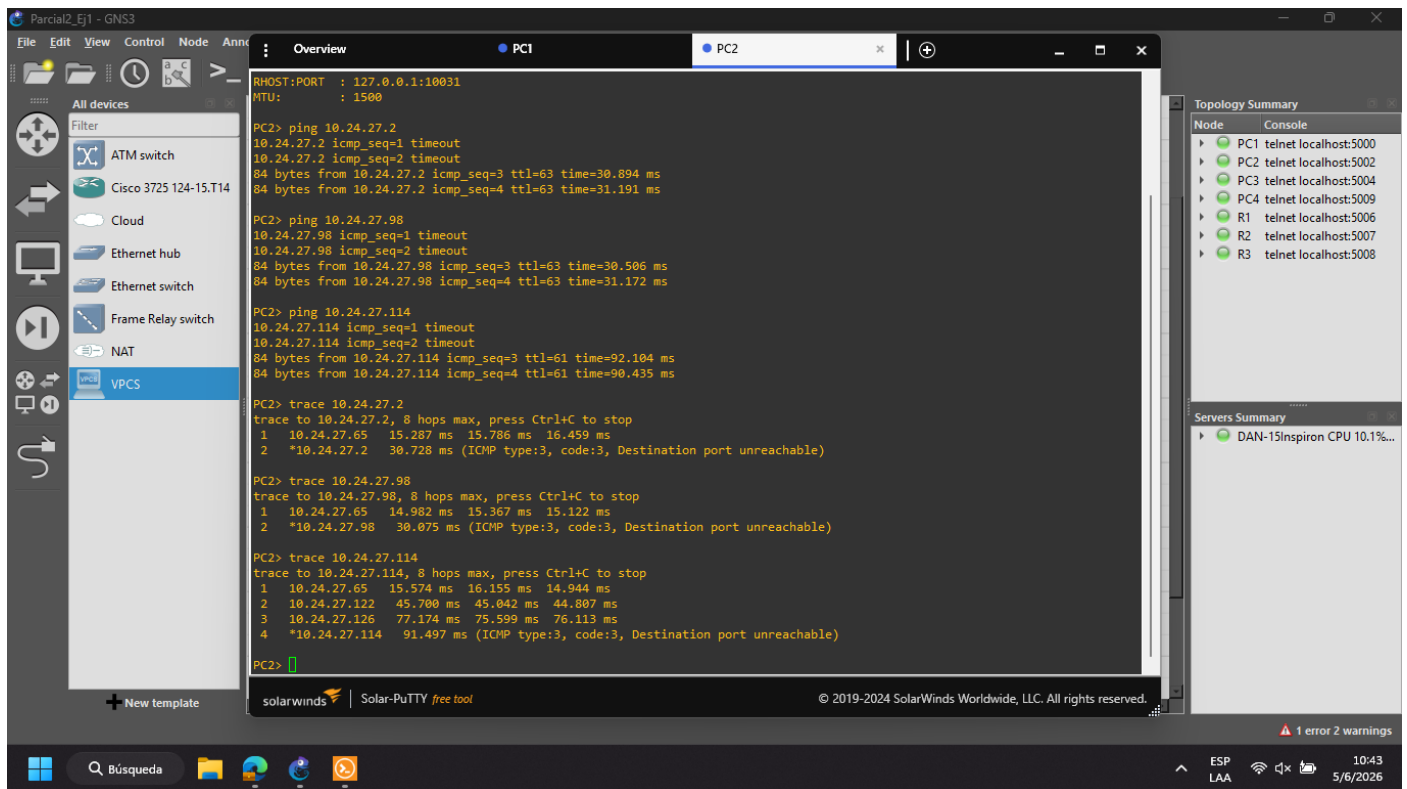
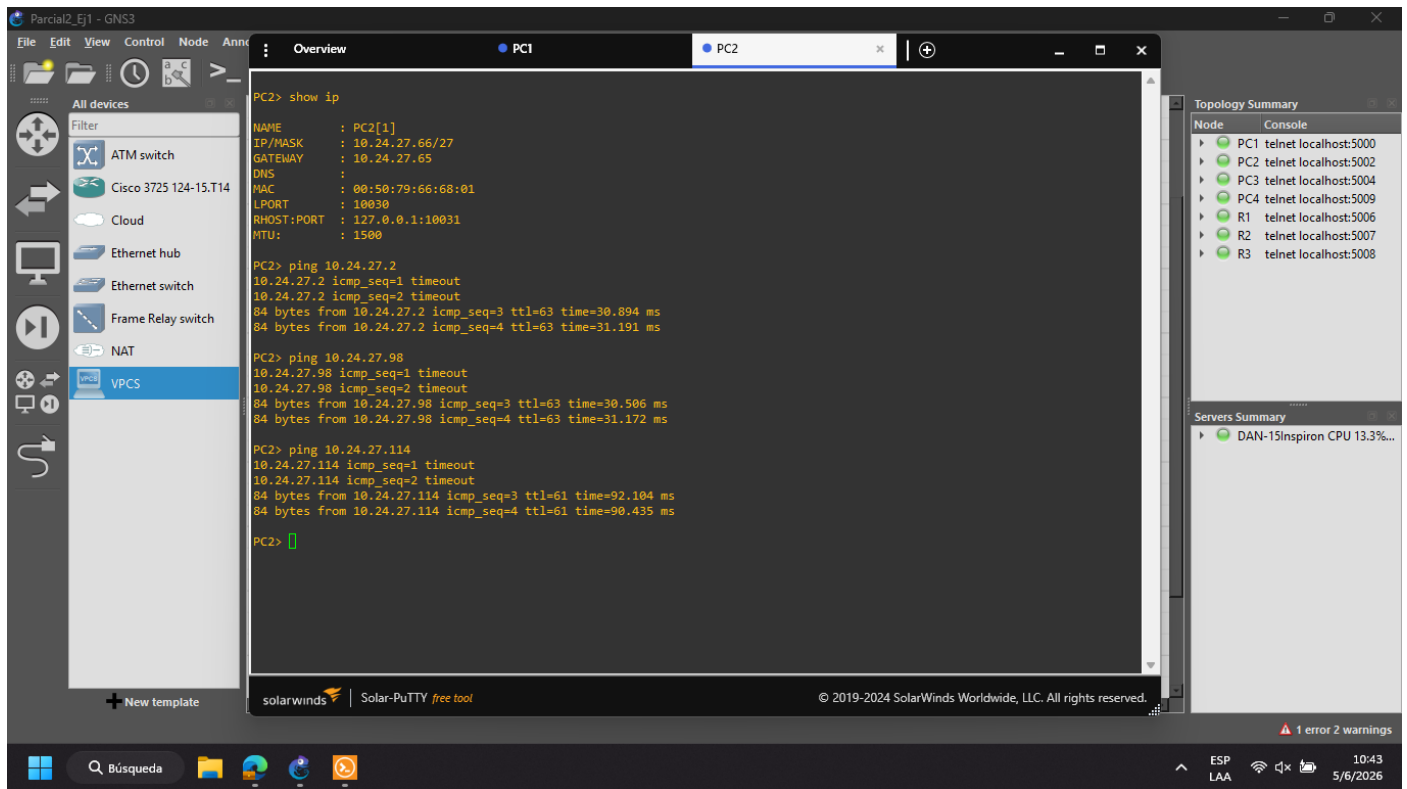
solarwinds | Solar-PuTTY free tool

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X: 0.0 Y: 0.0 Z: 0.5

1 error 2 warnings

10:41 5/6/2026



Parcial2_Ej1 - GNS3

File Edit View Control Node Annotations

Overview PC1 PC2 PC3

PC3> show ip

```
NAME       : PC3[1]
IP/MASK    : 10.24.27.98/28
GATEWAY    : 10.24.27.97
DNS        :
MAC        : 00:50:79:66:68:02
LPORT     : 10028
RHOST:PORT : 127.0.0.1:10029
MTU       : 1500
```

PC3> ping 10.24.27.2

```
84 bytes from 10.24.27.2 icmp_seq=1 ttl=63 time=29.917 ms
84 bytes from 10.24.27.2 icmp_seq=2 ttl=63 time=31.147 ms
84 bytes from 10.24.27.2 icmp_seq=3 ttl=63 time=32.059 ms
```

PC3> ping 10.24.27.66

```
84 bytes from 10.24.27.66 icmp_seq=1 ttl=63 time=31.935 ms
84 bytes from 10.24.27.66 icmp_seq=2 ttl=63 time=31.035 ms
84 bytes from 10.24.27.66 icmp_seq=3 ttl=63 time=30.540 ms
```

PC3> ping 10.24.27.114

```
84 bytes from 10.24.27.114 icmp_seq=1 ttl=61 time=90.110 ms
84 bytes from 10.24.27.114 icmp_seq=2 ttl=61 time=91.843 ms
84 bytes from 10.24.27.114 icmp_seq=3 ttl=61 time=91.611 ms
```

PC3> []

Topology Summary

Node	Console
PC1	telnet localhost:5000
PC2	telnet localhost:5002
PC3	telnet localhost:5004
PC4	telnet localhost:5009
R1	telnet localhost:5006
R2	telnet localhost:5007
R3	telnet localhost:5008

Servers Summary

DAN-15Inspiron CPU 8.8%, ...

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X: -33.0 Y: -22.0 Z: 1.0

1 error 2 warnings

ESP LAA 10:44 5/6/2026

Parcial2_Ej1 - GNS3

File Edit View Control Node Annotations

Overview PC1 PC2 PC3

RHOST:PORT : 127.0.0.1:10029
MTU: : 1500

PC3> ping 10.24.27.2

```
84 bytes from 10.24.27.2 icmp_seq=1 ttl=63 time=29.917 ms
84 bytes from 10.24.27.2 icmp_seq=2 ttl=63 time=31.147 ms
84 bytes from 10.24.27.2 icmp_seq=3 ttl=63 time=32.059 ms
```

PC3> ping 10.24.27.66

```
84 bytes from 10.24.27.66 icmp_seq=1 ttl=63 time=31.935 ms
84 bytes from 10.24.27.66 icmp_seq=2 ttl=63 time=31.035 ms
84 bytes from 10.24.27.66 icmp_seq=3 ttl=63 time=30.540 ms
```

PC3> ping 10.24.27.114

```
84 bytes from 10.24.27.114 icmp_seq=1 ttl=61 time=90.110 ms
84 bytes from 10.24.27.114 icmp_seq=2 ttl=61 time=91.843 ms
84 bytes from 10.24.27.114 icmp_seq=3 ttl=61 time=91.611 ms
```

PC3> trace 10.24.27.2

```
trace to 10.24.27.2, 8 hops max, press Ctrl+C to stop
 1  10.24.27.97  15.389 ms  15.353 ms  15.987 ms
 2  * * *
 3  *10.24.27.2  30.251 ms (ICMP type:3, code:3, Destination port unreachable)
```

PC3> trace 10.24.27.66

```
trace to 10.24.27.66, 8 hops max, press Ctrl+C to stop
 1  10.24.27.97  15.673 ms  14.069 ms  15.000 ms
 2  * * *
 3  *10.24.27.66  30.764 ms (ICMP type:3, code:3, Destination port unreachable)
```

PC3> trace 10.24.27.114

```
trace to 10.24.27.114, 8 hops max, press Ctrl+C to stop
 1  10.24.27.97  14.872 ms  16.085 ms  15.594 ms
 2  10.24.27.122  46.392 ms  45.010 ms  45.288 ms
 3  10.24.27.126  77.084 ms  75.153 ms  76.768 ms
 4  * * *
 5  *10.24.27.114  92.807 ms (ICMP type:3, code:3, Destination port unreachable)
```

PC3> []

Topology Summary

Node	Console
PC1	telnet localhost:5000
PC2	telnet localhost:5002
PC3	telnet localhost:5004
PC4	telnet localhost:5009
R1	telnet localhost:5006
R2	telnet localhost:5007
R3	telnet localhost:5008

Servers Summary

DAN-15Inspiron CPU 9.3%, ...

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X: -33.0 Y: -22.0 Z: 1.0

1 error 2 warnings

ESP LAA 10:45 5/6/2026

Parcial2_Ej1 - GNS3

File Edit View Control Node Annotations

Overview PC1 PC2 PC3 PC4

All devices

Filter

- ATM switch
- Cisco 3725 124-15.T14
- Cloud
- Ethernet hub
- Ethernet switch
- Frame Relay switch
- NAT
- VPCS

+ New template

X: 0.0 Y: 0.0 Z: 0.5

PC4> show ip

```
NAME       : PC4[1]
IP/MASK    : 10.24.27.114/29
GATEWAY    : 10.24.27.113
DNS        :
MAC        : 00:50:79:66:68:03
LPORT      : 10034
RHOST:PORT : 127.0.0.1:10035
MTU        : 1500
```

PC4> ping 10.24.27.2

```
84 bytes from 10.24.27.2 icmp_seq=1 ttl=61 time=93.227 ms
84 bytes from 10.24.27.2 icmp_seq=2 ttl=61 time=91.528 ms
```

PC4> ping 10.24.27.66

```
84 bytes from 10.24.27.66 icmp_seq=1 ttl=61 time=91.222 ms
84 bytes from 10.24.27.66 icmp_seq=2 ttl=61 time=92.853 ms
```

PC4> ping 10.24.27.98

```
84 bytes from 10.24.27.98 icmp_seq=1 ttl=61 time=91.085 ms
84 bytes from 10.24.27.98 icmp_seq=2 ttl=61 time=92.291 ms
```

PC4> []

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Topology Summary

Node	Console
PC1	telnet localhost:5000
PC2	telnet localhost:5002
PC3	telnet localhost:5004
PC4	telnet localhost:5009
R1	telnet localhost:5006
R2	telnet localhost:5007
R3	telnet localhost:5008

Servers Summary

- DAN-15Inspiron CPU 10.6%...

1 error 2 warnings

ESP LAA 10:46 5/6/2026

Parcial2_Ej1 - GNS3

File Edit View Control Node Annotations

Overview PC1 PC2 PC3 PC4

All devices

Filter

- ATM switch
- Cisco 3725 124-15.T14
- Cloud
- Ethernet hub
- Ethernet switch
- Frame Relay switch
- NAT
- VPCS

+ New template

X: 0.0 Y: 0.0 Z: 0.5

PC4> ping 10.24.27.2

```
84 bytes from 10.24.27.2 icmp_seq=1 ttl=61 time=93.227 ms
84 bytes from 10.24.27.2 icmp_seq=2 ttl=61 time=91.528 ms
```

PC4> ping 10.24.27.66

```
84 bytes from 10.24.27.66 icmp_seq=1 ttl=61 time=91.222 ms
84 bytes from 10.24.27.66 icmp_seq=2 ttl=61 time=92.853 ms
```

PC4> ping 10.24.27.98

```
84 bytes from 10.24.27.98 icmp_seq=1 ttl=61 time=91.085 ms
84 bytes from 10.24.27.98 icmp_seq=2 ttl=61 time=92.291 ms
```

PC4> trace 10.24.27.2

```
trace to 10.24.27.2, 8 hops max, press Ctrl+C to stop
 1 10.24.27.113 16.264 ms 15.084 ms 16.164 ms
 2 10.24.27.125 45.585 ms 46.827 ms 45.189 ms
 3 10.24.27.121 76.317 ms 75.900 ms 76.485 ms
 4 *10.24.27.2 90.636 ms (ICMP type:3, code:3, Destination port unreachable)
```

PC4> trace 10.24.27.66

```
trace to 10.24.27.66, 8 hops max, press Ctrl+C to stop
 1 10.24.27.113 15.716 ms 15.020 ms 15.582 ms
 2 10.24.27.125 44.959 ms 45.206 ms 47.101 ms
 3 10.24.27.121 76.058 ms 76.220 ms 74.905 ms
 4 *10.24.27.66 92.478 ms (ICMP type:3, code:3, Destination port unreachable)
```

PC4> trace 10.24.27.98

```
trace to 10.24.27.98, 8 hops max, press Ctrl+C to stop
 1 10.24.27.113 15.658 ms 15.447 ms 15.649 ms
 2 10.24.27.125 45.430 ms 45.433 ms 46.423 ms
 3 10.24.27.121 75.487 ms 77.825 ms 76.382 ms
 4 * * *
 5 *10.24.27.98 105.481 ms (ICMP type:3, code:3, Destination port unreachable)
```

PC4> []

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Topology Summary

Node	Console
PC1	telnet localhost:5000
PC2	telnet localhost:5002
PC3	telnet localhost:5004
PC4	telnet localhost:5009
R1	telnet localhost:5006
R2	telnet localhost:5007
R3	telnet localhost:5008

Servers Summary

- DAN-15Inspiron CPU 8.7%...

1 error 2 warnings

ESP LAA 10:47 5/6/2026

Ejercicio 2

Configuración de interfaces y tablas de rutas en routers:

The screenshot shows the GNS3 interface configuration for Router R1. The left sidebar displays the 'All devices' list with 'VPCS' selected. The main console window shows the configuration process for R1, including the IP address assignment for FastEthernet0/0, FastEthernet0/1, and FastEthernet1/0. The right sidebar shows the 'Topology Summary' and 'Servers Summary'.

Router R1 Configuration:

```
Building configuration...
[OK]
R1#
*Mar 1 00:06:47.271: %SYS-5-CONFIG_I: Configured from console by console
R1#
*Mar 1 00:13:19.955: %OSPF-5-ADJCHG: Process 1, Nbr 192.24.27.157 on FastEthernet
et1/0 from LOADING to FULL, Loading Done
R1#
*Mar 1 00:19:04.807: %OSPF-5-ADJCHG: Process 1, Nbr 192.24.27.149 on FastEthernet
et0/1 from LOADING to FULL, Loading Done
R1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.24.27.113   YES manual up          up
FastEthernet0/1 192.24.27.137   YES manual up          up
FastEthernet1/0 192.24.27.141   YES manual up          up
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

192.24.27.0/24 is variably subnetted, 11 subnets, 5 masks
O   192.24.27.128/29 [110/30] via 192.24.27.138, 00:05:57, FastEthernet0/1
C   192.24.27.136/30 is directly connected, FastEthernet0/1
C   192.24.27.140/30 is directly connected, FastEthernet1/0
O   192.24.27.144/30 [110/65] via 192.24.27.142, 00:12:06, FastEthernet1/0
O   192.24.27.148/30 [110/20] via 192.24.27.138, 00:10:02, FastEthernet0/1
O   192.24.27.152/30 [110/21] via 192.24.27.142, 00:02:01, FastEthernet1/0
O   192.24.27.156/30 [110/11] via 192.24.27.142, 00:15:53, FastEthernet1/0
O   192.24.27.64/27 [110/11] via 192.24.27.142, 00:16:22, FastEthernet1/0
O   192.24.27.96/28 [110/20] via 192.24.27.138, 00:10:37, FastEthernet0/1
C   192.24.27.112/28 is directly connected, FastEthernet0/0
O   192.24.27.0/26 [110/21] via 192.24.27.142, 00:02:03, FastEthernet1/0
R1#
```

The screenshot shows the GNS3 interface configuration for Router R2. The left sidebar displays the 'All devices' list with 'VPCS' selected. The main console window shows the configuration process for R2, including the IP address assignment for FastEthernet0/0, FastEthernet0/1, FastEthernet1/0, and Serial2/0. The right sidebar shows the 'Topology Summary' and 'Servers Summary'.

Router R2 Configuration:

```
*Mar 1 00:16:20.315: %OSPF-5-ADJCHG: Process 1, Nbr 192.24.27.149 on Serial2/0
from LOADING to FULL, Loading Done
R2#
*Mar 1 00:24:38.267: %OSPF-5-ADJCHG: Process 1, Nbr 192.24.27.158 on FastEthernet
et0/1 from LOADING to FULL, Loading Done
R2#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.24.27.65     YES manual up          up
FastEthernet0/1 192.24.27.157   YES manual up          up
FastEthernet1/0 192.24.27.142   YES manual up          up
Serial2/0       192.24.27.145   YES manual up          up
Serial2/2       unassigned      YES unset  administratively down down
Serial2/3       unassigned      YES unset  administratively down down
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

192.24.27.0/24 is variably subnetted, 11 subnets, 5 masks
O   192.24.27.128/29 [110/30] via 192.24.27.158, 00:02:39, FastEthernet0/1
O   192.24.27.136/30 [110/11] via 192.24.27.141, 00:16:54, FastEthernet1/0
C   192.24.27.140/30 is directly connected, FastEthernet1/0
C   192.24.27.144/30 is directly connected, Serial2/0
O   192.24.27.148/30 [110/21] via 192.24.27.158, 00:02:39, FastEthernet0/1
O   192.24.27.152/30 [110/20] via 192.24.27.141, 00:10:43, FastEthernet1/0
O   192.24.27.156/30 is directly connected, FastEthernet0/1
C   192.24.27.64/27 is directly connected, FastEthernet0/0
O   192.24.27.96/28 [110/21] via 192.24.27.141, 00:11:05, FastEthernet1/0
O   192.24.27.112/28 [110/11] via 192.24.27.141, 00:16:56, FastEthernet1/0
O   192.24.27.0/26 [110/20] via 192.24.27.158, 00:02:41, FastEthernet0/1
R2#
```

Parcial2_Ej2 - GNS3

File Edit View Control Node Annotations

All devices

Filter

ATM switch

Cisco 3725 124-15.T14

Cloud

Ethernet hub

Ethernet switch

Frame Relay switch

NAT

VPCS

New template

X: 0.0 Y: 0.0 Z: 0.5

R1 R2 R3

```
[OK]
R3#
*Mar 1 00:12:41.535: %SYS-5-CONFIG_I: Configured from console by console
R3#
*Mar 1 00:16:33.371: %OSPF-5-ADJCHG: Process 1, Nbr 192.24.27.153 on FastEthernet
et0/1 from LOADING to FULL, Loading Done
R3#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 192.24.27.97 YES manual up up
FastEthernet0/1 192.24.27.149 YES manual up up
FastEthernet1/0 192.24.27.138 YES manual up up
Serial2/0 unassigned YES unset administratively down down
Serial2/2 unassigned YES unset administratively down down
Serial2/3 unassigned YES unset administratively down down
R3#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

192.24.27.0/24 is variably subnetted, 11 subnets, 5 masks
O 192.24.27.128/29 [110/20] via 192.24.27.150, 00:07:15, FastEthernet0/1
C 192.24.27.136/30 is directly connected, FastEthernet1/0
O 192.24.27.140/30 [110/2] via 192.24.27.137, 00:11:38, FastEthernet1/0
C 192.24.27.144/30 is directly connected, Serial2/0
C 192.24.27.148/30 is directly connected, FastEthernet0/1
O 192.24.27.152/30 [110/20] via 192.24.27.150, 00:07:05, FastEthernet0/1
O 192.24.27.156/30 [110/12] via 192.24.27.137, 00:11:40, FastEthernet1/0
O 192.24.27.64/27 [110/12] via 192.24.27.137, 00:11:40, FastEthernet1/0
C 192.24.27.96/28 is directly connected, FastEthernet0/0
O 192.24.27.112/28 [110/11] via 192.24.27.137, 00:11:40, FastEthernet1/0
O 192.24.27.0/26 [110/22] via 192.24.27.137, 00:03:12, FastEthernet1/0
R3#
```

solarwinds | Solar-PuTTY free tool

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Topology Summary

Node Console

- PC1 telnet localhost:5003
- PC2 telnet localhost:5005
- PC3 telnet localhost:5007
- PC4 telnet localhost:5011
- PC6 telnet localhost:5013
- R1 telnet localhost:5000
- R2 telnet localhost:5001
- R3 telnet localhost:5002
- R4 telnet localhost:5009
- R5 telnet localhost:5010

Servers Summary

- DAN-15Inspiron CPU 9.9%...

ESP LAA

11:51 5/6/2026

1 error 3 warnings

Parcial2_Ej2 - GNS3

File Edit View Control Node Annotations

All devices

Filter

ATM switch

Cisco 3725 124-15.T14

Cloud

Ethernet hub

Ethernet switch

Frame Relay switch

NAT

VPCS

New template

X: 0.0 Y: 0.0 Z: 0.5

R1 R2 R3 R4

```
R4(config-router)#exit
R4(config)#end
R4#
building configuration...
[OK]
R4#
*Mar 1 00:13:32.123: %SYS-5-CONFIG_I: Configured from console by console
R4#
*Mar 1 00:17:06.219: %OSPF-5-ADJCHG: Process 1, Nbr 192.24.27.158 on FastEthernet
et0/1 from LOADING to FULL, Loading Done
R4#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 192.24.27.129 YES manual up up
FastEthernet0/1 192.24.27.153 YES manual up up
FastEthernet1/0 192.24.27.150 YES manual up up
R4#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

192.24.27.0/24 is variably subnetted, 11 subnets, 5 masks
C 192.24.27.128/29 is directly connected, FastEthernet0/0
O 192.24.27.136/30 [110/2] via 192.24.27.149, 00:07:45, FastEthernet1/0
O 192.24.27.140/30 [110/3] via 192.24.27.149, 00:07:45, FastEthernet1/0
O 192.24.27.144/30 [110/65] via 192.24.27.149, 00:07:45, FastEthernet1/0
C 192.24.27.148/30 is directly connected, FastEthernet1/0
C 192.24.27.152/30 is directly connected, FastEthernet0/1
O 192.24.27.156/30 [110/11] via 192.24.27.154, 00:03:42, FastEthernet0/1
O 192.24.27.64/27 [110/13] via 192.24.27.149, 00:07:47, FastEthernet1/0
O 192.24.27.96/28 [110/11] via 192.24.27.149, 00:07:47, FastEthernet1/0
O 192.24.27.112/28 [110/12] via 192.24.27.149, 00:07:47, FastEthernet1/0
O 192.24.27.0/26 [110/20] via 192.24.27.154, 00:04:02, FastEthernet0/1
R4#
```

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Topology Summary

Node Console

- PC1 telnet localhost:5003
- PC2 telnet localhost:5005
- PC3 telnet localhost:5007
- PC4 telnet localhost:5011
- PC6 telnet localhost:5013
- R1 telnet localhost:5000
- R2 telnet localhost:5001
- R3 telnet localhost:5002
- R4 telnet localhost:5009
- R5 telnet localhost:5010

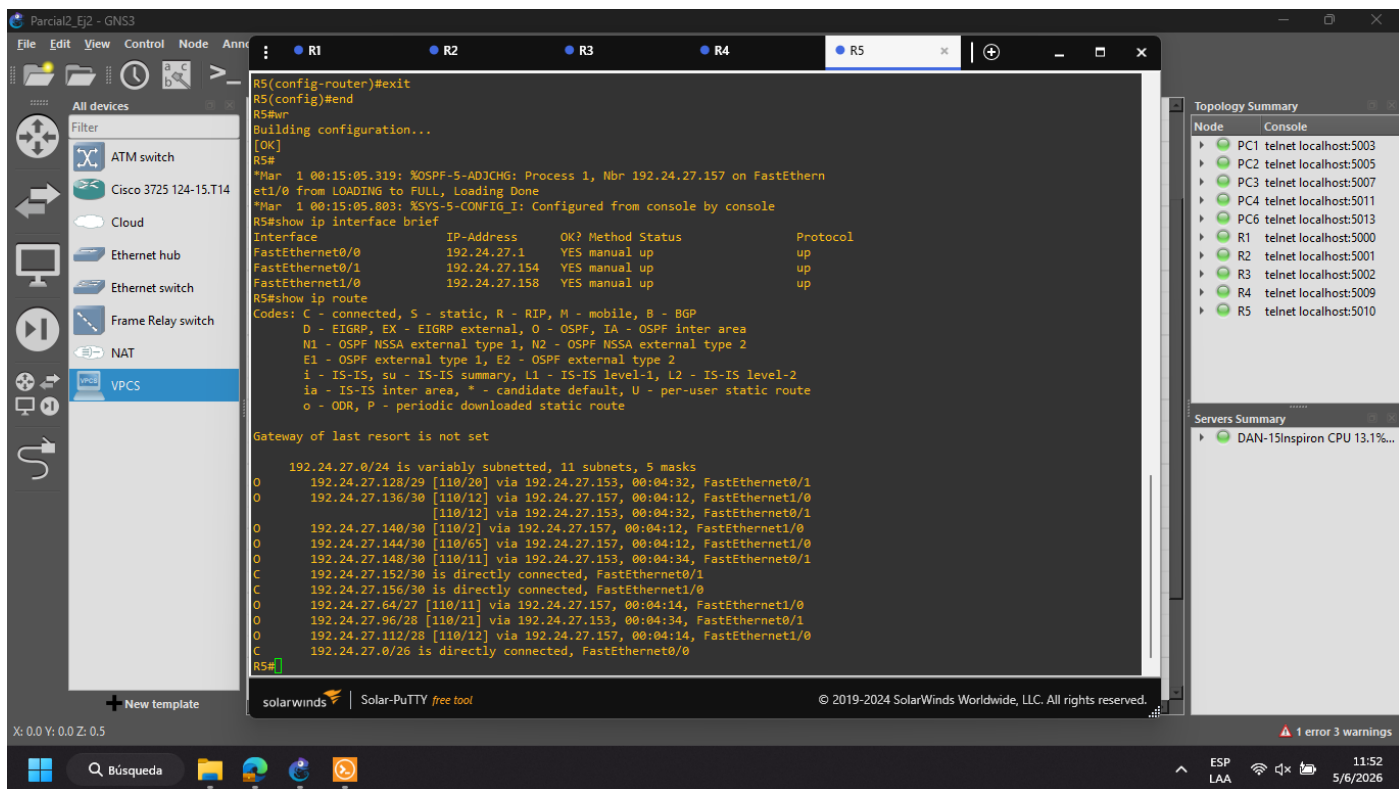
Servers Summary

- DAN-15Inspiron CPU 16.8%...

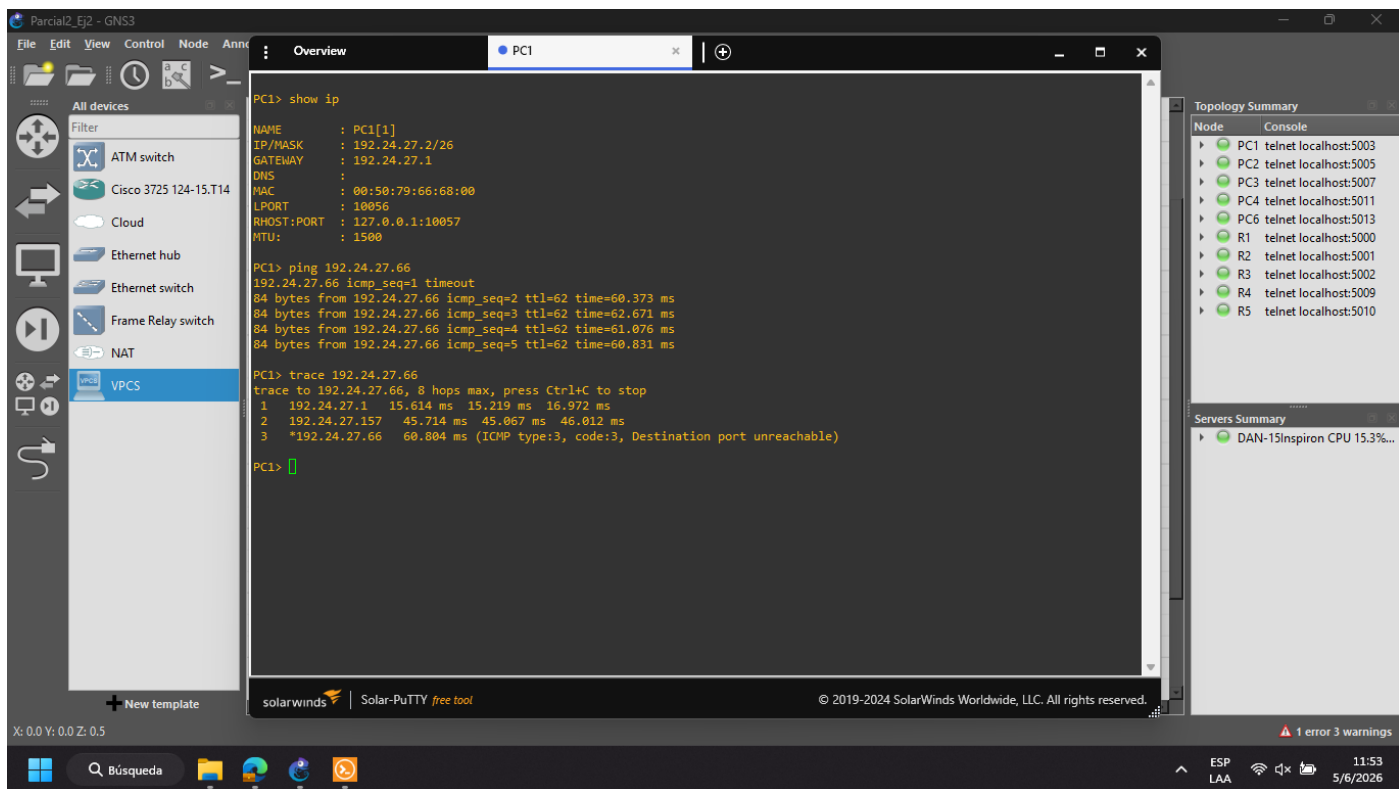
ESP LAA

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1 error 3 warnings



Configuración de IPs y pruebas de ping/trace entre PCs:



Parcial2_Ej2 - GNS3

File Edit View Control Node Annotations

Overview PC1 PC2

All devices

Filter

- ATM switch
- Cisco 3725 124-15.T14
- Cloud
- Ethernet hub
- Ethernet switch
- Frame Relay switch
- NAT
- VPCS

+ New template

X: 0.0 Y: 0.0 Z: 0.5

PC2> show ip

```
NAME       : PC2[1]
IP/MASK    : 192.24.27.66/27
GATEWAY    : 192.24.27.65
DNS        :
MAC        : 00:50:79:66:68:01
LPORT     : 10058
RHOST:PORT : 127.0.0.1:10059
MTU        : 1500
```

PC2> ping 192.24.27.98

```
192.24.27.98 icmp_seq=1 timeout
84 bytes from 192.24.27.98 icmp_seq=2 ttl=61 time=91.383 ms
84 bytes from 192.24.27.98 icmp_seq=3 ttl=61 time=92.477 ms
84 bytes from 192.24.27.98 icmp_seq=4 ttl=61 time=92.833 ms
84 bytes from 192.24.27.98 icmp_seq=5 ttl=61 time=92.006 ms
```

PC2> trace 192.24.27.98

```
trace to 192.24.27.98, 8 hops max, press Ctrl+C to stop
 1 192.24.27.65 15.443 ms 15.192 ms 15.497 ms
 2 192.24.27.141 46.851 ms 45.794 ms 45.766 ms
 3 192.24.27.138 77.437 ms 75.649 ms 76.497 ms
 4 *192.24.27.98 90.732 ms (ICMP type:3, code:3, Destination port unreachable)
```

PC2> []

solarwinds | Solar-PuTTY free tool

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Topology Summary

Node	Console
PC1	telnet localhost:5003
PC2	telnet localhost:5005
PC3	telnet localhost:5007
PC4	telnet localhost:5011
PC6	telnet localhost:5013
R1	telnet localhost:5000
R2	telnet localhost:5001
R3	telnet localhost:5002
R4	telnet localhost:5009
R5	telnet localhost:5010

Servers Summary

- DAN-15Inspiron CPU 12.4%...

1 error 3 warnings

ESP LAA

11:54 5/6/2026

Parcial2_Ej2 - GNS3

File Edit View Control Node Annotations

Overview PC1 PC2 PC3

All devices

Filter

- ATM switch
- Cisco 3725 124-15.T14
- Cloud
- Ethernet hub
- Ethernet switch
- Frame Relay switch
- NAT
- VPCS

+ New template

X: -107.5 Y: 45.5 Z: 1.0

PC3> show ip

```
NAME       : PC3[1]
IP/MASK    : 192.24.27.98/28
GATEWAY    : 192.24.27.97
DNS        :
MAC        : 00:50:79:66:68:02
LPORT     : 10060
RHOST:PORT : 127.0.0.1:10061
MTU        : 1500
```

PC3> ping 192.24.27.114

```
192.24.27.114 icmp_seq=1 timeout
84 bytes from 192.24.27.114 icmp_seq=2 ttl=62 time=60.885 ms
84 bytes from 192.24.27.114 icmp_seq=3 ttl=62 time=60.804 ms
84 bytes from 192.24.27.114 icmp_seq=4 ttl=62 time=59.989 ms
84 bytes from 192.24.27.114 icmp_seq=5 ttl=62 time=60.936 ms
```

PC3> trace 192.24.27.114

```
trace to 192.24.27.114, 8 hops max, press Ctrl+C to stop
 1 192.24.27.97 15.848 ms 15.568 ms 14.501 ms
 2 192.24.27.137 45.824 ms 45.812 ms 46.642 ms
 3 *192.24.27.114 60.603 ms (ICMP type:3, code:3, Destination port unreachable)
```

PC3> []

solarwinds | Solar-PuTTY free tool

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Topology Summary

Node	Console
PC1	telnet localhost:5003
PC2	telnet localhost:5005
PC3	telnet localhost:5007
PC4	telnet localhost:5011
PC6	telnet localhost:5013
R1	telnet localhost:5000
R2	telnet localhost:5001
R3	telnet localhost:5002
R4	telnet localhost:5009
R5	telnet localhost:5010

Servers Summary

- DAN-15Inspiron CPU 22.1%...

1 error 3 warnings

ESP LAA

11:55 5/6/2026

Parcial2_Ej2 - GNS3

File Edit View Control Node Annotations

Overview PC1 PC2 PC3 PC4

All devices

Filter

- ATM switch
- Cisco 3725 124-15.T14
- Cloud
- Ethernet hub
- Ethernet switch
- Frame Relay switch
- NAT
- VPCS

+ New template

X: 0.0 Y: 0.0 Z: 0.5

PC4> show ip

```
NAME       : PC4[1]
IP/MASK    : 192.24.27.114/28
GATEWAY    : 192.24.27.113
DNS        :
MAC        : 00:50:79:66:68:03
LPORT     : 10062
RHOST:PORT : 127.0.0.1:10063
MTU        : 1500
```

PC4> ping 192.24.27.130

```
192.24.27.130 icmp_seq=1 timeout
84 bytes from 192.24.27.130 icmp_seq=2 ttl=61 time=91.173 ms
84 bytes from 192.24.27.130 icmp_seq=3 ttl=61 time=93.204 ms
84 bytes from 192.24.27.130 icmp_seq=4 ttl=61 time=91.242 ms
84 bytes from 192.24.27.130 icmp_seq=5 ttl=61 time=90.607 ms
```

PC4> trace 192.24.27.130

```
trace to 192.24.27.130, 8 hops max, press Ctrl+C to stop
 1  192.24.27.113    15.655 ms  15.601 ms  15.217 ms
 2  192.24.27.138    45.460 ms  47.099 ms  45.480 ms
 3  192.24.27.150    75.248 ms  76.153 ms  75.698 ms
 4  *192.24.27.130   91.961 ms  (ICMP type:3, code:3, Destination port unreachable)
```

PC4> []

solarwinds | Solar-PuTTY free tool

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Topology Summary

Node	Console
PC1	telnet localhost:5003
PC2	telnet localhost:5005
PC3	telnet localhost:5007
PC4	telnet localhost:5011
PC6	telnet localhost:5013
R1	telnet localhost:5000
R2	telnet localhost:5001
R3	telnet localhost:5002
R4	telnet localhost:5009
R5	telnet localhost:5010

Servers Summary

- DAN-15Inspiron CPU 15.8%...

1 error 3 warnings

ESP LAA

11:56 5/6/2026

Parcial2_Ej2 - GNS3

File Edit View Control Node Annotations

Overview PC1 PC2 PC3 PC4 PC6

All devices

Filter

- ATM switch
- Cisco 3725 124-15.T14
- Cloud
- Ethernet hub
- Ethernet switch
- Frame Relay switch
- NAT
- VPCS

+ New template

X: -107.5 Y: 45.5 Z: 1.0

PC6> show ip

```
NAME       : PC6[1]
IP/MASK    : 192.24.27.130/29
GATEWAY    : 192.24.27.129
DNS        :
MAC        : 00:50:79:66:68:04
LPORT     : 10064
RHOST:PORT : 127.0.0.1:10065
MTU        : 1500
```

PC6> ping 192.24.27.2

```
192.24.27.2 icmp_seq=1 timeout
192.24.27.2 icmp_seq=2 timeout
84 bytes from 192.24.27.2 icmp_seq=3 ttl=62 time=60.484 ms
84 bytes from 192.24.27.2 icmp_seq=4 ttl=62 time=60.907 ms
84 bytes from 192.24.27.2 icmp_seq=5 ttl=62 time=60.356 ms
```

PC6> trace 192.24.27.2

```
trace to 192.24.27.2, 8 hops max, press Ctrl+C to stop
 1  192.24.27.129    14.809 ms  15.872 ms  15.043 ms
 2  192.24.27.154    46.670 ms  45.957 ms  45.040 ms
 3  *192.24.27.2     61.831 ms  (ICMP type:3, code:3, Destination port unreachable)
```

PC6> []

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Topology Summary

Node	Console
PC1	telnet localhost:5003
PC2	telnet localhost:5005
PC3	telnet localhost:5007
PC4	telnet localhost:5011
PC6	telnet localhost:5013
R1	telnet localhost:5000
R2	telnet localhost:5001
R3	telnet localhost:5002
R4	telnet localhost:5009
R5	telnet localhost:5010

Servers Summary

- DAN-15Inspiron CPU 10.9%...

1 error 3 warnings

ESP LAA

11:57 5/6/2026